RMO 2002

Exercise 1 (RMO 2002 P2,). Solve the following equation for real x:

$$(x^2+x-2)^3+(2x^2-x-1)^3=27(x^2-1)^3.$$

Solution 1. The given equation is equivalent to

$$(x^2+x-2)^3+(2x^2-x-1)^3+(-3x^2+3)^3=0.$$

Note that $x^2+x-2, 2x^2-x-1, -3x^2+3$ add up to zero. This implies

$$\begin{aligned}&(x^2+x-2)^3+(2x^2-x-1)^3+(-3x^2+3)^3\\&=3(x^2+x-2)(2x^2-x-1)(-3x^2+3)\\&=-9(x+2)(x-1)(x-1)(2x-1)(x-1)(x+1).\end{aligned}$$

Thus the required solutions for x are

$$-2,-1,\frac{1}{2},1.$$

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Exercise 2 (RMO 2002 P3, , cf. RMO 2012d P2, RMO 2012a P2, RMO 2012b P2, RMO 2012c P2). Let a,b,c be positive integers such that a divides b^2, b divides c^2 and c divides a^2. Prove that abc divides $(a+b+c)^7$.

Solution 2. In the following, k denotes one of the integers $2,3,4,5$. Assume that $a \mid b^k, b \mid c^k, c \mid a^k$. Note that if one of a,b,c is equal to 1, then all of them are equal to 1, and then there is nothing to prove. Henceforth, let us assume that a,b,c are larger than 1. The given conditions imply that the set of prime divisors of a,b,c are equal.

Let p be a common prime divisor of a,b,c. Let $p^\alpha, p^\beta, p^\gamma$ denote the highest powers of p dividing a,b,c respectively. The given divisibility conditions imply that

$$\alpha \le k\beta, \beta \le k\gamma, \gamma \le k\alpha.$$

This yields

$$\alpha+\beta+\gamma \le \alpha + k^2\alpha + 2\alpha = (k^2+k+1)\alpha,$$

and similarly, $\alpha+\beta+\gamma \le (k^2+k+1)\beta$, $\alpha+\beta+\gamma \le (k^2+k+1)\gamma$, which shows that

$$\alpha+\beta+\gamma \le (k^2+k+1)\min\{\alpha,\beta,\gamma\}.$$

Note that $p^{\min\{\alpha,\beta,\gamma\}}$ divides $a+b+c$. By the above inequality, $p^{\alpha+\beta+\gamma}$ divides $(a+b+c)^{k^2+k+1}$. So abc divides $(a+b+c)^{k^2+k+1}$. Also note that $p^{\alpha+\beta+\gamma}$ is the highest power of p dividing abc. This shows that for each common prime divisor p of a,b,c, the highest power of p that divides abc also divides $(a+b+c)^{k^2+k+1}$. It follows that abc divides $(a+b+c)^{k^2+k+1}$. ■

Solution 3. It suffices to show that abc divides $a^\alpha b^\beta c^\gamma$ for any nonnegative integers α,β,γ such that $\alpha+\beta+\gamma = k^2+k+1$. Note that

$$abc \mid ac^{k+1} \mid a^{k^2+k+1}, \quad abc \mid a^{k+1}b \mid b^{k^2+k+1}, \quad abc \mid b^{k+1}c \mid c^{k^2+k+1},$$

which shows that abc divides $a^{k^2+k+1}, b^{k^2+k+1}, c^{k^2+k+1}$. Moreover, abc divides $a^\alpha b^\beta c^\gamma$ when α,β,γ are all positive. It remains to consider the case when exactly one of α,β,γ is zero, which we assume from now on. If $\alpha = 0$, then

$$abc \mid b^{k+1}c \mid b^\beta c^\gamma \text{ if } \beta \ge k+1,$$
$$abc \mid c^{k^2+1}b \mid b^\beta c^\gamma \text{ if } \beta \le k$$

holds. If $\beta = 0$, then

$$abc \mid c^{k+1}a \mid c^\gamma a^\alpha \text{ if } \gamma \ge k+1,$$
$$abc \mid a^{k^2+1}c \mid c^\gamma a^\alpha \text{ if } \gamma \le k$$

holds. If $\gamma = 0$, then

$$abc \mid a^{k+1}b \mid a^\alpha b^\beta \text{ if } \alpha \ge k+1,$$
$$abc \mid b^{k^2+1}a \mid a^\alpha b^\beta \text{ if } \alpha \le k$$

holds. This shows that abc divides $(a+b+c)^{k^2+k+1}$. ■

Exercise 3 (RMO 2002 P4,). Suppose the integers $1,2,3,\ldots,10$ are split into two disjoint collections a_1,a_2,a_3,a_4,a_5 and b_1,b_2,b_3,b_4,b_5 such that $a_1<a_2<a_3<a_4<a_5$ and $b_1>b_2>b_3>b_4>b_5$.

(i) Show that the larger number in any pair $\{a_j,b_j\}$, $1\le j\le 5$, is at least 6.

(ii) Show that $|a_1-b_1|+|a_2-b_2|+|a_3-b_3|+|a_4-b_4|+|a_5-b_5| = 25$ for every such partition.

The content posted here and at this blog by Evan Chen are quite useful.

Solution 4. Let $1 \leq j \leq 5$ be an integer. Note that the inequalities

$$a_1 < a_2 < \cdots < a_j, \quad b_j > b_{j+1} > \cdots > b_5$$

hold. Since $a_1, a_2, \ldots, a_j, b_j, b_{j+1}, \ldots, b_5$ are six distincts elements of the set $\{1, 2, \ldots, 10\}$, their maximum is at least 6. The above inequalities show that this maximum is equal to $\max\{a_j, b_j\}$. This proves part (i).

Note that the sets $\{a_1, b_1\}, \{a_2, b_2\}, \{a_3, b_3\}, \{a_4, b_4\}, \{a_5, b_5\}$ are pairwise disjoint subsets of $\{1, 2, \ldots, 10\}$, and the maximum of any of these sets is at least 6. Consequently, the maximums of these sets are equal to $6, 7, 8, 9, 10$ in some order. Thus their minimums are equal to $1, 2, 3, 4, 5$ in some order. Hence, the integer

$$|a_1 - b_1| + |a_2 - b_2| + |a_3 - b_3| + |a_4 - b_4| + |a_5 - b_5|$$

is equal to the difference of the sum of of the maximums of these sets and the sum of the minimums of these sets, which is equal to

$$6 + 7 + 8 + 9 + 10 - (1 + 2 + 3 + 4 + 5) = 25.$$

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Exercise 4 (RMO 2002 P6,). For any natural number $n > 1$, prove the inequality

$$\frac{1}{2} < \frac{1}{n^2+1} + \frac{2}{n^2+2} + \frac{3}{n^2+3} + \cdots + \frac{n}{n^2+n} < \frac{1}{2} + \frac{1}{2n}.$$

Solution 5. Note that

$$\begin{aligned}
&\frac{1}{n^2+1} + \frac{2}{n^2+2} + \frac{3}{n^2+3} + \cdots + \frac{n}{n^2+n} \\
&> \frac{1}{n^2+n}(1 + 2 + 3 + \cdots + n) \quad (\text{using } n > 1) \\
&= \frac{1}{2}.
\end{aligned}$$

Also note that

$$\begin{aligned}
&\frac{1}{n^2+1} + \frac{2}{n^2+2} + \frac{3}{n^2+3} + \cdots + \frac{n}{n^2+n} \\
&< \frac{1}{n^2+1}(1 + 2 + 3 + \cdots + n) \quad (\text{using } n > 1) \\
&= \frac{n^2+n}{2(n^2+1)} \\
&= \frac{1}{2} + \frac{n-1}{2(n^2+1)} \\
&< \frac{1}{2} + \frac{1}{2n}.
\end{aligned}$$

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Exercise 5 (RMO 2002 P7,). Find all integers a, b, c, d satisfying the following relations:

1. $1 \leq a \leq b \leq c \leq d$;
2. $ab + cd = a + b + c + d + 3$.

Solution 6. Let a, b, c, d be integers satisfying the given conditions. Let p, q, r be integers satisfying $b = a + p, c = b + q, d = c + r$. Note that p, q, r are nonnegative. The given conditions yield

$$a(a + p) + (a + p + q)(a + p + q + r) = 4a + 3p + 2q + r + 3,$$

which is equivalent to

$$2a^2 + (3p + 2q + r)a + (p + q)(p + q + r) = 4a + 3p + 2q + r + 3 = 0,$$

that is,

$$2a(a - 2) + (a - 1)(3p + 2q + r) + (p + q)(p + q + r) = 3. \tag{1}$$

From Eq. (1), we get $a \leq 2$.

If $a = 1$ holds, then $(p + q)(p + q + r) = 5$, which gives

$$p + q = 1, p + q + r = 5,$$

that is, (p, q, r) is equal to $(1, 0, 4)$ or $(0, 1, 4)$.

Let us consider the case that $a = 2$. Note that

$$(3p + 2q + r) + (p + q)(p + q + r) = 3,$$

which implies that $p = 0$. It follows that $2q + r + q(q + r) = 3$, which gives $q \leq 1$, and hence (q, r) is equal to $(0, 3)$ or $(1, 0)$.

The above argument shows that (p, q, r) is equal to one of

$$(1, 0, 4), (0, 1, 4), (0, 0, 3), (0, 1, 0).$$

Consequently, (a, b, c, d) is equal to one of

$$(1, 2, 2, 6), (1, 1, 2, 6), (2, 2, 2, 5), (2, 2, 3, 3).$$

Moreover, the above tuples also satisfy the given conditions.

We conclude that the above tuples are precisely all the solutions to the given relations. ■